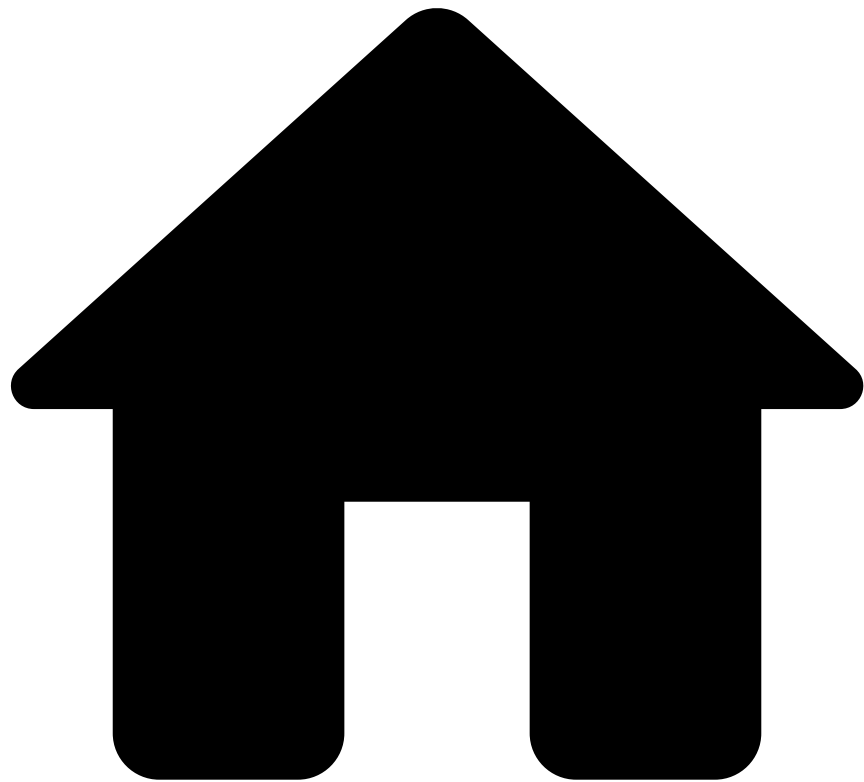


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Configuring Kerberos for Data Science Resources

Pulse stores data science resources on the filesystem. To work with those resources in Pulse, users need to have access to the respective files. You can use group properties to grant access.

This page describes how to use groups and group properties to grant a group access to the following data science resources:

- Data sources (created/used by plan executions, model evaluations, rule evaluations, sandbox evaluations)
- Statistic file of data source analysis
- Model binaries from model training and hyperparameter tuning
- Spark archive file (initialization of a leader app jvm process)
- Spark-libs (initialization of a leader app jvm process)

Hierarchical groups and access rights

When Pulse users try to access a data science resource, Pulse needs to access the respective files using Kerberos credentials. These credentials can be defined either in the Group Properties field of a group, or in the *pulse.properties* file.

Pulse doesn't only consider the groups that are explicitly assigned to the user, but also their parent groups in the group hierarchy.

If none of the user's groups or their parent groups has any Kerberos credentials defined for the given resource, Pulse tries to fall back on credentials defined in the *pulse.properties* file, if there are any.

Group properties

To configure a group to use specific Kerberos credentials, add the following key-value pairs in the **Group Properties** field of the group:

Key	Value
defaultPath	The path to the folder of the tenant on the HDFS.
kerberos.keytabPath	The path to the keytab file of the tenant.
kerberos.principal	The principal authentication to be used for Kerberos. Normally, it has the following format: <code>username/resource@REALM</code> .

Using groups to control visibility

The members of the groups that are set up this way can control who can see resources stored in the group's directory.

To set the resource visible only to that specific group, complete the following steps. Note that the steps presume that the user has the membership of the group(s) that they select in step 3.

1. Open the resource.
2. Select the Advanced Options button, and select **Specific Groups**. Select the group from the list of groups.
3. In the list of permissions, select **Read / Update / Delete** according to your choice. A check mark appears next to them when selected.
4. **Save** the resource.

If the visibility of a data science resource is public (i.e. not limited to a specific group), then Pulse will use the Kerberos credentials defined in the *pulse.properties* file, if there are any.

Example

Let's consider an example where you want to configure the following Pulse groups to access HDFS directories:

- *TeamA* and *TeamB* are two groups that should have access only to their own respective directories, plus a directory with common resources (containing Spark archives and Spark libs).
- The members of the *landlords* group should have access to all directories. They also have a directory for their own resources that should not be accessed by anyone else.

To configure this, the following needs to be set up:

1. HDFS directories:

Directory	Description	Permissions	Owner	Group
/user/ pulse_common/	The common directory that stores files common to all teams, for example spark-libs, spark-archive, etc. The user of each team must belong to its group to be able to access this directory.	770	pulse_common	pulse_common

Directory	Description	Permissions	Owner	Group
/user/ pulse_landlord/	This directory is specific to the landlord. It is used to store files with sensitive data of all teams.	770	pulse_landlord	pulse_landlord
/user/teamA/	This directory is specific to TeamA. It stores the data science resources of this team.	770	pulse_teamA	pulse_teamA
/user/teamB/	This directory is specific to TeamB. It stores the data science resources of this team.	770	pulse_teamB	pulse_teamB

2. HDFS users:

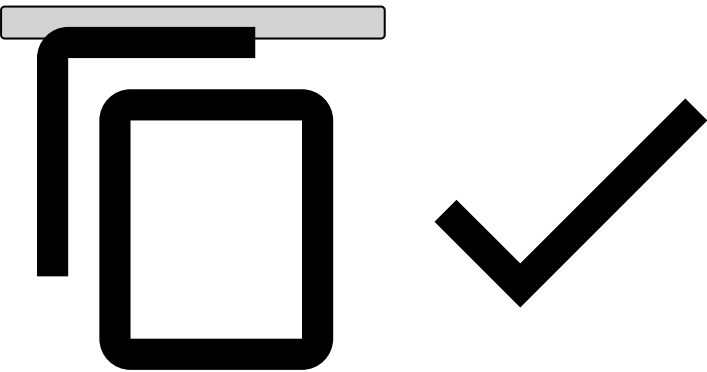
Each of these users must be configured with Kerberos authentication using a keytab file.

User name	Description	Member of the group(s)	Maps to Pulse group
pulse_common	The HDFS user through which Pulse users will access common files.	pulse_common	all groups in Pulse
pulse_landlord	The HDFS user through which the landlords will access landlord-specific files.	pulse_landlord, pulse_common, pulse_teamA, pulse_teamB	landlords
pulse_teamA	The HDFS user through which members of TeamA will access their own files.	pulse_teamA, pulse_common	TeamA
pulse_teamB	The HDFS user through which members of TeamB will access their own files.	pulse_teamB, pulse_common	TeamB

3. Pulse properties in the *pulse.properties* file:

Replace the placeholders (marked with `<...>`) with the respective values. These properties allow Pulse users to access common files.

```
pulse.global.services.modules.distributed.jobs.kerberos.enabled=true
pulse.global.services.modules.distributed.jobs.kerberos.principal=<principal_of_pulse_common>
pulse.global.services.modules.distributed.jobs.kerberos.keytabPath=<keytab_path_of_pulse_common>
pulse.global.services.modules.distributed.jobs.spark.master=yarn-client
pulse.global.services.modules.distributed.clusterDirToReceive=hdfs:///user/pulse_common/spark-libs
```



4. Pulse groups and users:

In Pulse, create three groups:

- landlords
- TeamA
- TeamB

For each of these groups, add the following key-value pairs in the **Group Properties** field:

Key	Value
defaultPath	The path to the respective HDFS directory of the group.
kerberos.keytabPath	The path to the group's keytab file.
kerberos.principal	The principal authentication to be used for Kerberos. Normally, it has the following format: username/resource@REALM.

Add the users to their respective group. Landlord users must be added to the teams' groups as well.

After configuring the above, users must set the resources visible only to the group. If the visibility is not restricted to the respective group, Pulse will use the credentials that are defined in the *pulse.properties* file to access the resource.